

Course Code: CS3001	Course Name: Computer Networks
Instructor Names: Mr. Shoaib Raza, and Mr. Muhammad Ali	
Student Roll No:	Section No:

Instructions:

- Return the question paper. There are 4 Questions and written on 5 pages.
- **All questions must be answered in answer script and according to the sequence given in the question paper.**
- In case of any ambiguity, you may make an assumption. But your assumption should not contradict any statement in the question paper.

Allowed Time: 03 Hours

Max Points: 100 Points

Question # 1:

CLO-02

[04 * 05 = 20 Points]

Part (a)

Consider the Figure 01, Suppose a Client from Institutional network has a request rate from the browsers to origin server is 20 requests/seconds with average object size is 500 KB. Web cache is also installed, which has the Hit ratio is 67%. Calculate the LAN and Access link utilization.

Part (b)

From Figure 01, assume that the client wants to retrieve the www.nu.edu.pk home page but has no information about the www.nu.edu.pk web server IP address. When obtaining the IP address for the host name www.nu.edu.pk, assume the roundtrip time between local DNS server and DNS root server is 5RTT, between local DNS server and DNS TLD server is 3RTT, and between the clients and the local DNS server is RTT.

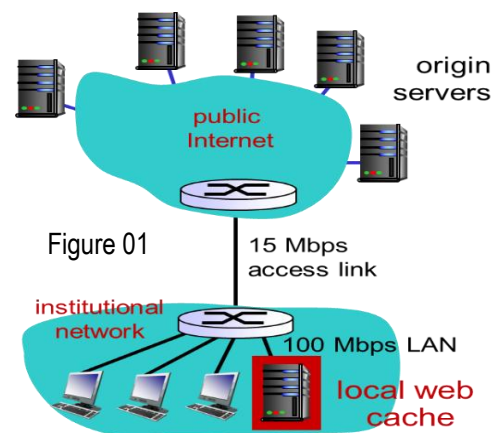
- How long does it take for the first client to obtain the IP address for www.nu.edu.pk?
- After that first client, a second client (on the same network) also wants to obtain the IP address for www.nu.edu.pk. How long for the second client?

Part (c)

It was observed during the Lab Final Exam that accessing the File storage has become a bottleneck due to multiple users trying to access the hosted lectures on the local drive. What application-level solution will be best suited to overcome this problem? Explain.

Part (d)

How DNS helps in identifying a server nearest to the user in a CDN setup? Explain with the help of a labelled diagram.

Question # 2:

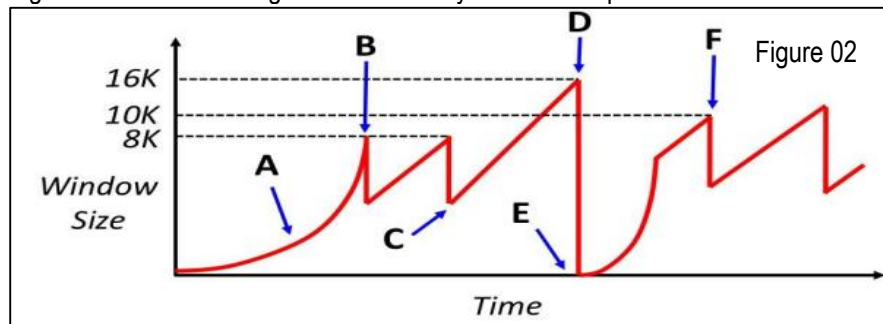
CLO-03

[2 * 10 = 20 Points]

Part (a)

Let's suppose a window size of 4 and sequence number range from 0 to 8. Apply both Go-Back-N (GBN) and Selective Repeat (SR) for sender and receiver events and actions. Assume that 2nd packet in the window is lost. Show the SR and GBN operations using the diagram in the presence of the lost packet.

Considering the following scenario shown in Figure 02 and briefly answer the questions below.



- Explain the event at B which occurs and causes the sender to decrease its window.
- Does the event at B necessitate that the network discarded a packet? Explain.
- Explain the event at D which occurs and causes the sender to decrease its window.
- Does the event at D necessitate that the network discarded a packet? Explain.
- What phenomenon event A represents? Explain.

Question # 3:

CLO-03

[05 * 04 = 20 Points]

Part (a)

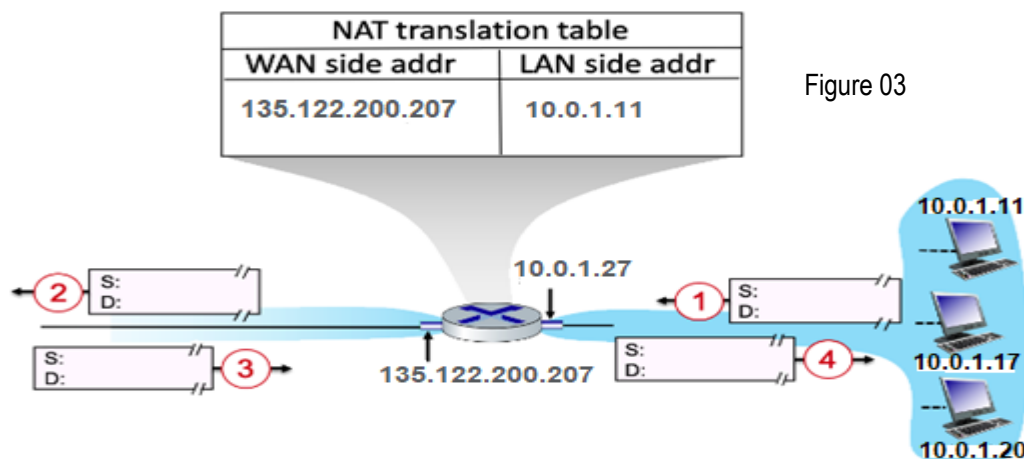
Suppose Ali wants to connect his laptop to his university network. Assume that the IT department of the University is running various services to support network operations e.g., DHCP, DNS, File storage, and web application (e.g., Flex, SLATE etc.) What sort of communication (types of messages) will be shared between Ali's Laptop and Server?

Part (b)

What four pieces of information does a DHCP client (Ali's Laptop) obtain from the DHCP server after a successful communication between the Client and Server?

Part (c)

Consider the scenario shown in Figure 03, in which three hosts, with private IP addresses 10.0.1.11, 10.0.1.17, 10.0.1.20 are in a local network behind a NAT'd router that sits between these three hosts and the larger Internet. IP datagrams being sent from, or destined to, these three hosts must pass through this NAT router. The router's interface on the LAN side has IP address 10.0.1.27, while the router's address on the Internet side has IP address 135.122.200.207

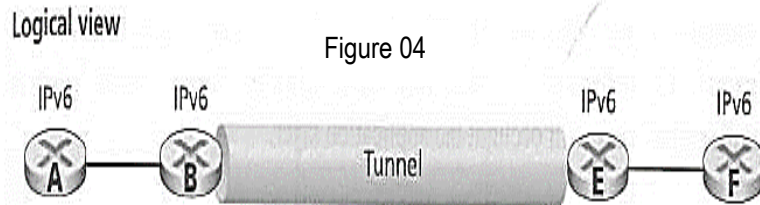


Suppose that the host with IP address 10.0.1.11 sends an IP datagram destined to host 128.119.176.181. The source port is 3405, and the destination port is 80.

- Consider the datagram at step 1, after it has been sent by the host but before it has reached the router. What is the source and destination IP address for this datagram?
- Now consider the datagram at step 2, after it has been transmitted by the router. What is the source and destination IP address for this datagram?
- At step 3, what is the source and destination IP address for this datagram?
- At step 4, what is the source and destination IP address for this datagram?

Part (d)

ISPs are slowly changing their IP networks to IPv6. This means they have to interconnect between groups of routers running IPv4 with other groups running IPv6. Suppose a scenario shown in Figure 04, where IPv6 packets need to traverse an IPv4 tunnel. Describe how this tunnel is practically implemented.



Part (e)

A packet is to be forwarded to a network with MTU of 1500 bytes. The packet has an IP header of 20 bytes and data part of 3980 bytes. Find number of fragments using the given table.

	Total Length	Flag	Fragment offset
Original Packet			
Fragment 1			
Fragment 2			
Fragment 3			
Fragment 4			

Question # 4:

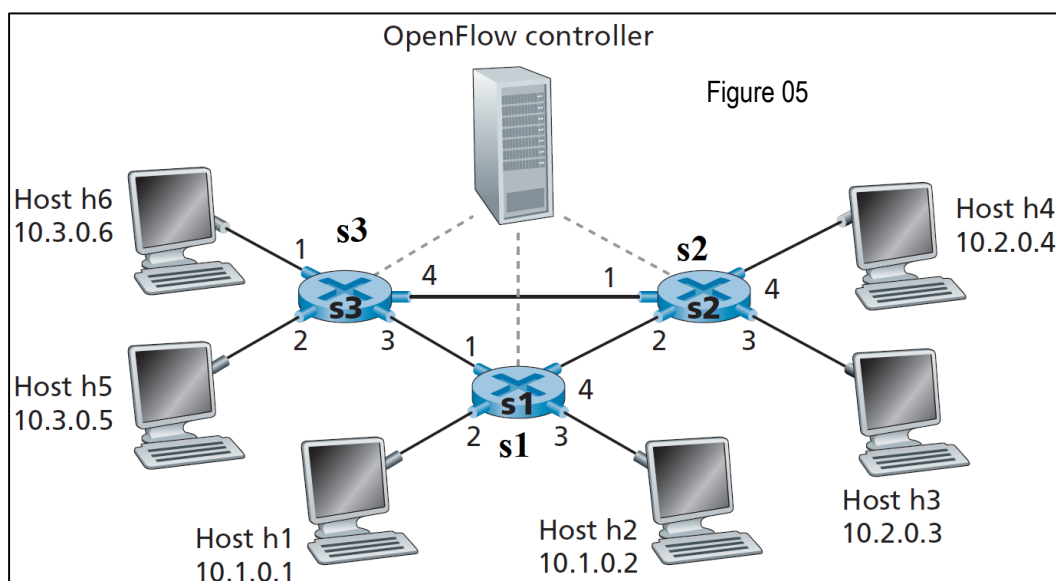
CLO-03

[05 * 08 = 40 Points]

Part (a)

For the network shown in Figure 05, suppose that the desired forwarding behavior is that packets from h5 or h6 destined to h3 or h4 are to be forwarded from s3 to s1, and then from s1 to s2 (thus completely avoiding the use of the link between s3 and s2).

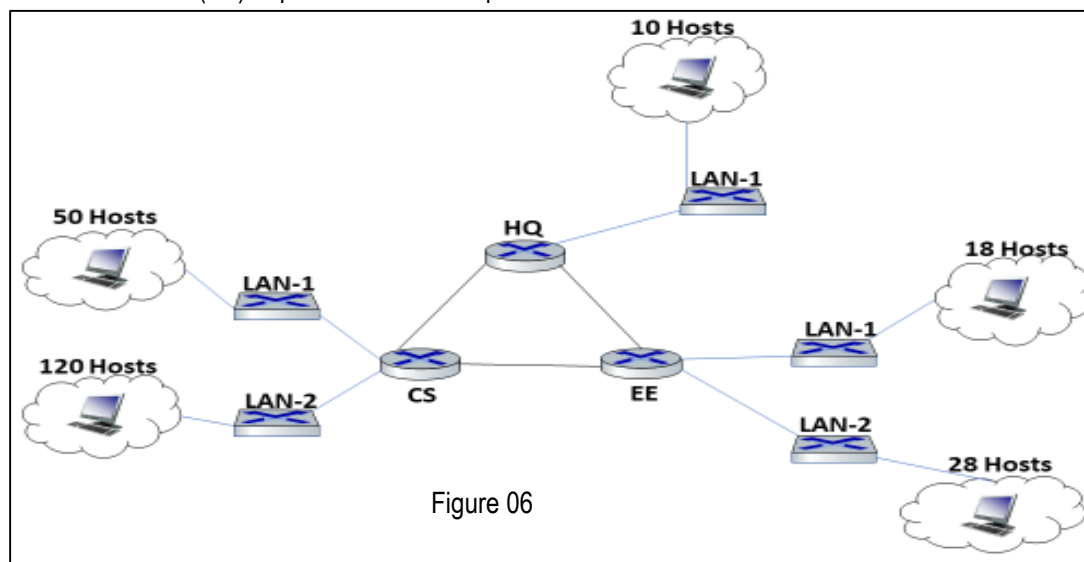
- Write down the flow table entry for s3, so that datagram sent from h5 or h6 are forwarded to s1 over interface 3.
- Write down the flow table entry for s1, so that datagram arriving at port 1 of s1, from s3, are forwarded to s2 over outgoing interface 4.
- Finally write down the flow table entry for s2, for forwarding to required destinations.



Part (b)

A University uses the address block of 192.168.9.0/24 for its network. Now your task is to assign IP addresses to the different department's LAN as shown in network Figure 06. The network has the following addressing requirements.

- The Computer Science (CS) department LAN-1 will require 50 host IP addresses.
- The Computer Science (CS) department LAN-2 will require 120 host IP addresses.
- The Electrical (EE) department LAN-1 will require 10 host IP addresses.
- The Electrical (EE) department LAN-2 require 18 host IP addresses.

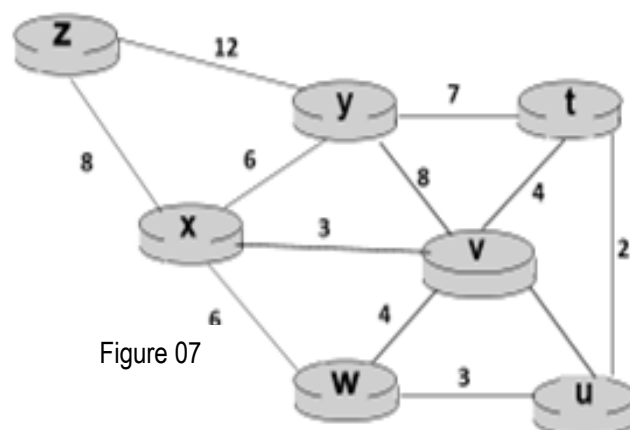


- Determine the following.
- Number of Bits Borrowed
- Total Number of Subnets
- Total Number of Host Addresses
- Number of Usable Addresses
- Custom Subnet Mask & Subnet Range (Network & Broadcast address)

Part (c)

Consider the graph given in Figure 07. With the indicated link costs, use Dijkstra's shortest-path algorithm to compute the shortest path from **node w** to all other nodes. Use the table given below for computations. Show the forwarding table for **node w**.

Node (N')	D(x), p(x)	D(u), p(u)	D(v), p(v)	D(t), p(t)	D(y), p(y)	D(z), p(z)
w						



Part (d)

A group of autonomous systems is shown in Figure 08. Answer the following questions:

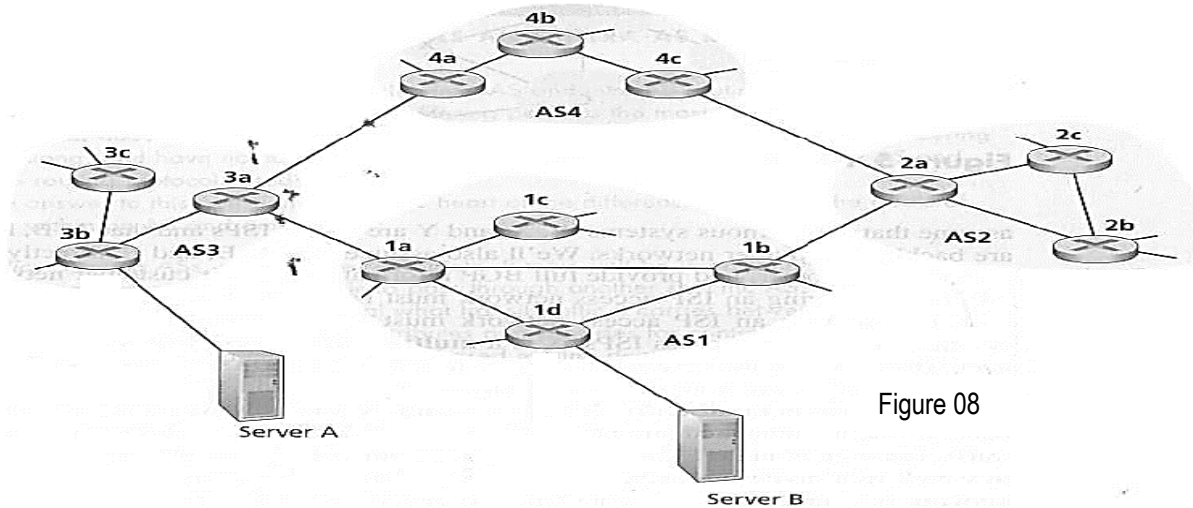


Figure 08

- Identify all BGP speaking routers in this network and label them appropriately as per your knowledge of BGP protocol.
- Using your knowledge of BGP protocol, how a PC attached to router 3b will reach Server B, if the link between router 3a and 1a is not operational.

Part (e) Consider the Figure 09 below and answer the following questions:

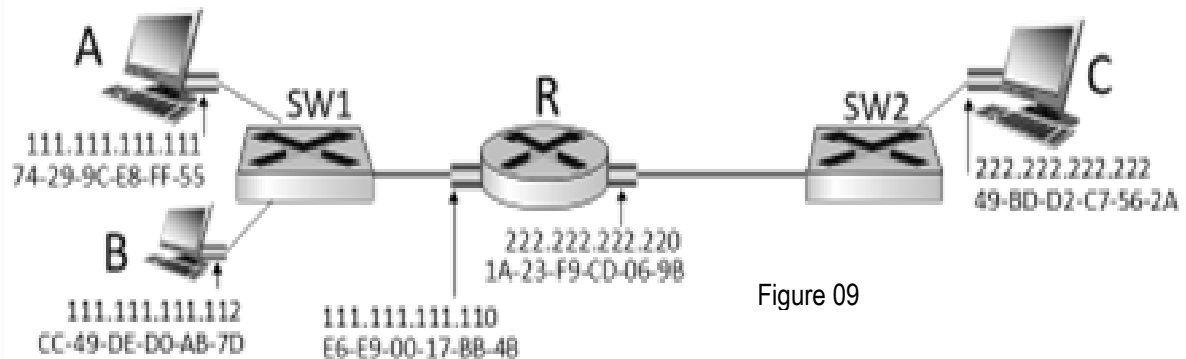


Figure 09

- Consider sending an IP datagram from PC-A to PC-B. PC-A issues an ARP request because it needs to send a packet to PC-B. In this scenario, what will happen next?
- Suppose PC-A would like to send IP datagram to PC-C on a remote LAN, however there are currently no mappings in the ARP cache. How will the device obtain a destination MAC address?
- What will be the content of the destination IP-address field in the header of the IP datagram sent by PC-A to PC-C?

Best of Luck!!!